

OpenNebula: A Free Solution for Building Clouds



OpenNebula is a free and open source software solution for building clouds and for data centre virtualisation. It is based on open technologies and is distributed under the Apache License 2. OpenNebula has features for scalability, integration, security and accounting. It offers cloud users and administrators a choice of interfaces.

OpenNebula is an open source platform for constructing virtualised private, public and hybrid clouds. It is a simple yet feature-rich, flexible solution to build and manage data centre virtualisation and enterprise clouds. So, with OpenNebula, virtual systems can be administered and monitored centrally on different Hyper-V and storage systems. When a component fails, OpenNebula takes care of the virtual instances on a different host system. The integration and automation of an existing heterogeneous landscape is highly flexible without further hardware investments.

Benefits of OpenNebula

The plurality of support to Hyper-V and platform-independent architecture makes OpenNebula the ideal solution for heterogeneous computing centre environments.

The main advantages of OpenNebula are:

- It is 100 per cent open source and offers all the features in one edition.
- It provides control via the command line or Web interface, which is ideal for a variety of user groups and needs.

- OpenNebula is available for all major Linux distributions, thus simplifying installation.
- The long-term use of OpenNebula in large scale production environments has proven its stability and flexibility.
- OpenNebula is interoperable and supports OCCI (Open Cloud Computing Interface) and AWS (Amazon Web Services).

Key features of OpenNebula

OpenNebula has features for scalability, integration, security and accounting. The developers also claim that it supports standardisation, interoperability and portability. It allows cloud users and administrators to choose from several cloud interfaces. Figure 1 shows the important features of OpenNebula.

Why OpenNebula?

- **Web interface or CLI - the choice is yours**
By using the OpenNebula CLI or Web interface, you can keep track of activities at any time. There is a central